

LunarVision: Technical Implementation Report

End-to-End Multi-Modal Lunar Image Registration & Temporal Change Detection Pipeline

Project: LunarVision	Status: Fully Implemented & Verified
Architecture: React 18 + FastAPI + OpenCV	Primary Detectors: SIFT & AKAZE
Scientific Ingestion: PDS3/PDS4 .IMG & Web Formats	Machine Learning: None (Deterministic Classical CV)

1. PROJECT OVERVIEW

LunarVision is an end-to-end computer vision platform engineered for planetary surface exploration and orbital image correspondence. It performs automated multi-modal image ingestion, contrast-aware CLAHE enhancement, scale- and rotation-invariant feature extraction (SIFT & AKAZE), Lowe's ratio-tested feature matching, RANSAC projective homography registration, quantitative alignment metric calculation, and temporal change isolation.

Main Objective: To provide sub-pixel geometric alignment and explainable temporal surface alteration detection across multi-temporal and multi-sensor lunar orbital passes (such as Chandrayaan OHRC, TMC, IIRS, and NASA LRO), eliminating false registration distortions while providing transparent scientific quality metrics.

Key Technologies: React 18, Vite, Tailwind CSS, FastAPI (Python 3.14 / 3.11+), OpenCV 4.13.0, NumPy, scikit-image 0.26.0, Pillow 12.1.1.

Database / Storage: Local stateless filesystem storage (``backend/outputs/`` and ``backend/uploads/``). No external database required.

2. SYSTEM ARCHITECTURE

The system follows a decoupled client-server architecture:

- **Frontend Layer** (``frontend/src/``): React 18 SPA serving interactive modules for Registration, Three-Sensor Analysis, and Change Detection.
- **API Gateway** (``backend/routers/image_routes.py``): FastAPI endpoint routing handling multipart FormData uploads and JSON responses.
- **Preprocessing Pipeline** (``backend/services/preprocessing.py``): PDS3/PDS4 label parsing, byte offset slicing, 16-bit to 8-bit dynamic range scaling, and CLAHE.
- **Feature Engine** (``backend/services/feature_detection.py``): SIFT feature extraction with automatic fallback to AKAZE.
- **Matching Engine** (``backend/services/matching.py``): Brute-Force KNN matcher with distance norm selection (L2 for SIFT, Hamming for AKAZE) and Lowe's Ratio Test.
- **Registration Engine** (``backend/services/registration.py``): RANSAC 2D perspective homography matrix estimation and bilinear warping.
- **Metrics Engine** (``backend/services/metrics.py``): Masked RMSE, Inlier Ratio, SSIM, and composite Confidence Score computation.
- **Change Engine** (``backend/services/change_detection.py``): SSIM dissimilarity, thresholding, morphological opening/closing, and connected component heuristic labeling.

3 & 4. FRONTEND & BACKEND IMPLEMENTATION

Endpoint	Method	Input Format	Purpose & Output
/api/health	GET	None	Returns telemetry: OpenCV, skimage, Python versions and directory status.
/api/images/register	POST	Multipart FormData (source, reference, detector)	Executes full SIFT/AKAZE registration, computes metrics & artifacts.
/api/images/multi-sensor-register	POST	Multipart FormData (hub, sensor_a, sensor_b)	Executes real pairwise cross-sensor registration across payloads.
/api/images/change-detection	POST	Multipart FormData (source, reference, min_conf, diff_thresh)	Executes registration followed by SSIM change detection & bounding box labels.

5 & 6. IMAGE PREPROCESSING & SCIENTIFIC .IMG SUPPORT

Scientific lunar datasets (NASA PDS3/PDS4, ISRO Chandrayaan TMC/OHRC/IIRS, LRO LROC) are delivered in .IMG formats containing ASCII text headers followed by 8-bit, 16-bit, or 32-bit float raw binary matrices. LunarVision implements an in-house PDS label parser in PreprocessingService.parse_pds_img():

- **Header Parsing:** Decodes the first 64KB as latin-1 and uses regular expressions to extract LINES, LINE_SAMPLES, SAMPLE_BITS, SAMPLE_TYPE, RECORD_BYTES, LABEL_RECORDS, and ^IMAGE byte pointer.
- **Endianness & Data Types:** Maps MSB (big-endian) and LSB (little-endian) types to NumPy dtypes (>u2, , >i2, , >f4,).
- **Percentile Normalization:** Applies 0.5%–99.5% percentile intensity scaling (np.percentile(img_matrix, (0.5, 99.5))) to convert high-dynamic range 16-bit or float arrays to 8-bit uint8 while preserving crater rims and regolith shadow contrast.
- **Raw Binary Fallback:** Unheadered .IMG files are decoded via parse_raw_binary_img() using square root factor reconstruction (side = int(np.sqrt(num_elems))).
- **CLAHE Contrast Enhancement:** Applies cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8)) to accentuate subtle topography.

7 – 11. FEATURE DETECTION, MATCHING, LOWE RATIO & RANSAC

Pipeline Stage	Algorithm / Library	Parameters / Config	Mathematical / Operational Role
Feature Detection	SIFT (OpenCV) <i>Fallback: AKAZE</i>	min_features=12 SIFT: DoG Pyramid AKAZE: M-LDB Binary	Extracts scale/rotation invariant keypoints. SIFT yields float32 descriptors; AKAZE yields uint8 binary descriptors.
Feature Matching	BFMatcher (OpenCV)	k=2 (KNN) SIFT: NORM_L2 AKAZE: NORM_HAMMING	Computes 2 nearest neighbors for each source descriptor against reference descriptor space.
Match Filtering	Lowe's Ratio Test	\$ au_{sift} = 0.75\$ \$ au_{akaze} = 0.80\$	Prunes ambiguous matches: Keeps match iff \$d_1 < \tau \times d_2\$. Discards duplicate terrain patterns.
Outlier Rejection	RANSAC (OpenCV)	reprojThresh=3.5px min_matches=6	Randomly samples 4-point subsets to compute \$3\times 3\$ Homography \$H\$, isolating geometrically consistent inliers.
Image Warping	Perspective Warp	BORDER_REFLECT INTER_LINEAR	Warp source image into reference coordinate space via \$p_{ref} \sim H p_{src}\$. Extends valid overlap mask.

12. SAME LOCATION DECISION LOGIC

The system enforces strict mathematical decision gates to avoid forcing false registrations on unrelated imagery:

```
def evaluate_same_location(ref_kps, src_kps, matches, inliers, confidence):  
    if len(ref_kps) < 6 or len(src_kps) < 6: return 'no_match'  
    if len(matches) < 6: return 'no_match'  
    if inliers < 6: return 'no_match'  
    if confidence < 0.20: return 'no_match'  
    return 'success'
```

14 & 15. QUANTITATIVE METRICS & FORMULAS

• **Masked RMSE (Root Mean Square Error):** Computes pixel intensity discrepancy over valid overlapping mask:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{(x,y) \in \text{Mask}} (I_{\text{ref}}(x,y) - I_{\text{aligned}}(x,y))^2}$$

• **Inlier Ratio:** Percentage of candidate matches surviving RANSAC outlier rejection:

$$\text{Inlier Ratio} = \frac{\text{Inlier Count}}{\max(\text{Good Matches}, 1)}$$

• **Registration Confidence Score:** Multi-factor synthesized reliability score:

$$\text{Confidence} = 0.40 \times \min\left(1.0, \frac{\text{Inliers}}{40}\right) + 0.35 \times \text{InlierRatio} + 0.25 \times \max\left(0.0, 1.0 - \frac{\text{RMSE}}{60}\right)$$

17 & 18. AUTOMATED TEST RESULTS & SAMPLE DATASETS

Test Script	Target Feature Tested	Sample Data Used	Empirical Results	Status
test_img_upload.py	PNG/JPG decoding, PDS3 16-bit .IMG, raw binary, SIFT/AKAZE & error rejection	01_baseline_pre_event.png 05_scientific_lunar_reference.IMG	6/6 tests passed. 276 inliers verified on PDS .IMG registration.	PASS
test_backend_cv.py	Full CV pipeline (Preprocess, SIFT, RANSAC, SSIM, No-match)	Synthetic crater terrain pair	RMSE: 7.83px, Inliers: 165/170, Confidence: 95.7%	PASS
test_api_endpoints.py	FastAPI endpoints (/health, /register, /change-detection)	Sample PNG orbital frames	HTTP 200 responses, static asset URL serving verified.	PASS
test_full_integration.py	End-to-end integration across multi-sensor channels & errors	OHRC, TMC, IIRS sample pairs	Multi-sensor Pair 1: 100 inliers (98.2% conf). Pair 2: 41 inliers (93.6% conf).	PASS

19 & 20. AI / ML USAGE & EXPLICIT CLARIFICATION

Explicit Clarification: No trained Machine Learning (ML), Deep Learning (DL), or Neural Network model is used in LunarVision. The entire system is grounded strictly in **deterministic classical computer vision algorithms** (gradient orientation histograms, non-linear scale space differential operators, and RANSAC geometric consensus). SIFT and AKAZE are hand-crafted mathematical feature extractors, not trained AI models.

24. CURRENT IMPLEMENTATION STATUS

Requirement	Status	Evidence File	Notes
JPEG/JPG/PNG Upload	IMPLEMENTED	PreprocessingService	Fully operational & tested.
Scientific .IMG PDS Decoding	IMPLEMENTED	preprocessing.py:L29	Parses PDS3/PDS4/VICAR labels & raw binary matrices.
SIFT Feature Detector	IMPLEMENTED	feature_detection.py:L77	Primary feature extractor (128d float32 descriptors).
AKAZE Feature Detector	IMPLEMENTED	feature_detection.py:L89	Fallback & binary feature extractor (Hamming norm).
Lowe's Ratio Test	IMPLEMENTED	matching.py:L73	Thresholds: 0.75 (SIFT) / 0.80 (AKAZE).
RANSAC Homography Warping	IMPLEMENTED	registration.py:L60	Tolerance: 3.5px, min matches: 6.
Same-Location Decision Logic	IMPLEMENTED	image_routes.py:L192	Rejects noise & bad pairs with 'no_match' alert.
Temporal Change Detection	IMPLEMENTED	change_detection.py	SSIM dissimilarity & heuristic bounding box tags.

26 & 27. DEMO READINESS & FINAL SUMMARY

Demo Readiness: YES. The system is fully functional, fully tested, and ready for live demonstration.

Final Summary for Technical Reviewers: LunarVision is an end-to-end computer vision platform for multi-modal lunar orbital imagery. Built with React 18, FastAPI, OpenCV, and scikit-image, it automatically ingests web imagery (PNG, JPG, TIFF, BMP) and scientific PDS3/PDS4 .IMG orbital datasets. The pipeline executes CLAHE contrast enhancement, scale-invariant SIFT and AKAZE keypoint extraction, Lowe's ratio-tested k-NN descriptor matching, and RANSAC perspective homography warping. The system computes quantitative alignment metrics (masked RMSE, inlier ratio, and confidence score) and performs SSIM-based temporal change detection to isolate surface anomalies. All operations are explainable, deterministic, and grounded in classical computer vision principles.